

Step 12 - Banking Domain Deep-Dive

Phase D · Regulators · Payment Rails · EOD Reality · ~20 min

Goal: The vocabulary layer - regulators in depth, Canadian payment rails, and EOD batch reality. Light reading + one artifact: docs/banking_domain_notes.md. This is what makes you sound like an insider in interviews.

LEARN 1 — The Regulators In Depth

Body	What it does	What it means for a DE
OSFI	Federal prudential regulator of banks, insurers, pensions. Guideline B-13 covers technology and cyber risk.	7-year retention, audit trails, data lineage, change management discipline, Delta time travel as the as-of answer
FINTRAC	AML and anti-terrorist-financing agency. Receives LCTR (cash >= \$10,000), EFTR (international EFT >= \$10,000), STR (suspicious, no threshold).	You build the detection pipelines and the report extracts (CSV out)
PIPEDA	Federal privacy law (Quebec adds Law 25). Consent, minimal collection, safeguards.	SIN/account masking, FSA-level postal codes, PII dropped at the Bronze-Silver boundary, access controls per layer
CDIC	Insures eligible deposits up to \$100,000 per category.	Deposit categorization data feeds CDIC reporting
FCAC	Consumer protection (fees, disclosure).	Complaint and fee analytics
Payments Canada	Owns and operates the national payment systems.	The rails your transaction data rides on - next section

LEARN 2 — Canadian Payment Rails (Interviewers Love This)

Rail	What it is	Typical data shape
Lynx	High-value RTGS (real-time gross settlement) system, replaced LVTS in 2021. Wires and large institutional payments.	Low volume, very high value, settlement-time critical
ACSS	Automated Clearing Settlement System - the batch retail rail: cheques, pre-authorized debits, EFT batches.	High volume, batch files, EOD heavy
Interac e-Transfer	Uniquely Canadian email/SMS-addressed transfer. Near-real-time.	JSON-ish event feeds, fraud-sensitive, FINTRAC-relevant
Interac Debit	POS debit network (not Visa/Mastercard).	High-volume card events with merchant data
RTR	Real-Time Rail - Canada's instant-payments system (phased rollout).	Streaming-friendly; banks are building event pipelines for it now
SWIFT wires	International transfers.	EFTR territory at >= \$10,000

One-liner to keep: "Lynx for high-value, ACSS for retail batch, Interac for person-to-person and POS, RTR coming for instant - and everything at or above \$10,000 crossing borders is EFTR territory."

LEARN 3 — EOD Batch Reality

The nightly rhythm at a Canadian bank: branches close, cut-off ~6-7 PM ET; source extracts land 7-11 PM; ingestion and Silver builds overnight; reconciliation gates Gold; reports and dashboards ready before 9 AM ET. Month-end and quarter-end runs are 2-3x volume - the reason your clusters autoscale and your pipelines parameterize by `business_date` for backfills.

Vocabulary that signals experience: EOD/BOD, cut-off, settlement date vs transaction date, GL, suspense account, reconciliation break, backfill, month-end spike, P1 incident, change freeze (banks freeze deployments around fiscal year-end and holidays).

LAB — Artifact: docs/banking_domain_notes.md

Write the notes file in your own words (30-40 lines). Required sections: the six bodies above with one personal-words line each; the rails table compressed; FINTRAC thresholds (LCTR/EFTR \$10,000, STR no threshold, structuring just-below pattern); an EOD timeline sketch; ten vocabulary terms with one-line definitions. Writing it yourself is the retention mechanism - do not paste mine.

CHECKPOINT

Self-test without notes: name the three reporting types and thresholds; explain Lynx vs ACSS in one sentence each; explain why postal codes get truncated to FSA; explain what a change freeze is. Phase D complete when the file is committed.